

智能巡检车 PRO 版实验手册

激光雷达 SLAM 建图实战实验

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1

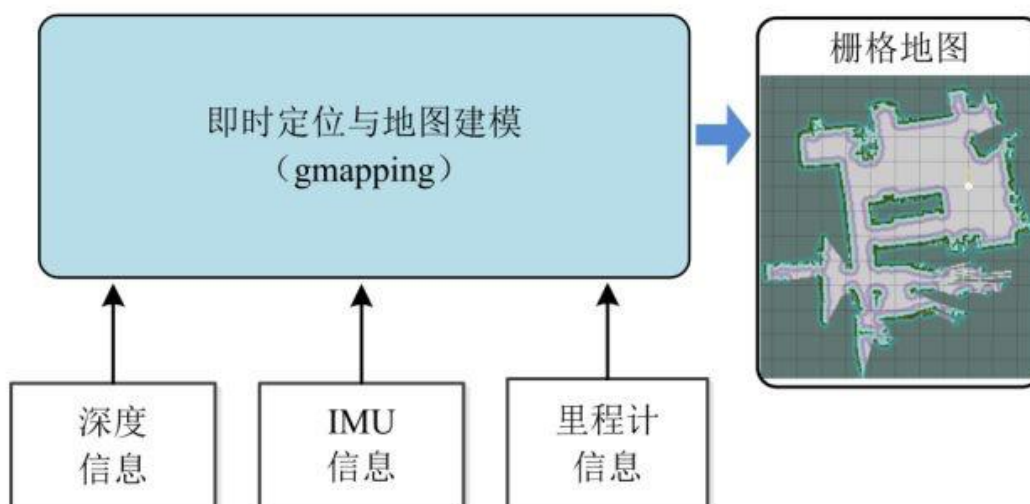
Gmapping 建图算法

1.1 简介

- gmapping 只适用于单帧二维激光点数小于 1440 的点,如果单帧激光点数大于 1440,那么就会出现 【[mapping-4] process has died】这样的问题。
- Gmapping 是基于滤波 SLAM 框架的常用开源 SLAM 算法。
- Gmapping 基于 RBpf 粒子滤波算法, 即时定位和建图过程分离, 先进行定位再进行建图。
- Gmapping 在 RBpf 算法上做了两个主要的改进: 改进提议分布和选择性重采样。

优点: Gmapping 可以实时构建室内地图, 在构建小场景地图所需的计算量较小且精度较高。

缺点: 随着场景增大所需的粒子增加, 因为每个粒子都携带一幅地图, 因此在构建大地图时所需内存和计算量都会增加。因此不适合构建大场景地图。并且没有回环检测, 因此在回环闭合时可能会造成地图错位, 虽然增加粒子数目可以使地图闭合但是以增加计算量和内存为代价。



1.2 使用

1.2.1 具体使用

注意：建图时，速度慢一些效果越好（注要是旋转速度要慢些），速度太快，效果会很差。

首先在宿主机【小车】中需要做端口绑定操作【参见端口绑定教程章节】，这里主要用到雷达和串口两个设备；

然后检查雷达和串口设备是否处于端口绑定状态：在宿主机【小车】参考如下命令执行查看，成功绑定是如下状态：

```
jetson@ubuntu:~$ ll /dev | grep ttyUSB*
lrwxrwxrwx 1 root root 7 Apr 21 14:52 myserial -> ttyUSB0
lrwxrwxrwx 1 root root 7 Apr 21 14:52 rplidar -> ttyUSB1
crwxrwxrwx 1 root dialout 188, 0 Apr 21 14:52 ttyUSB0
crwxrwxrwx 1 root dialout 188, 1 Apr 21 14:52 ttyUSB1
```

若显示未绑定雷达或者串口设备，可以插拔一下USB连接线再次查看

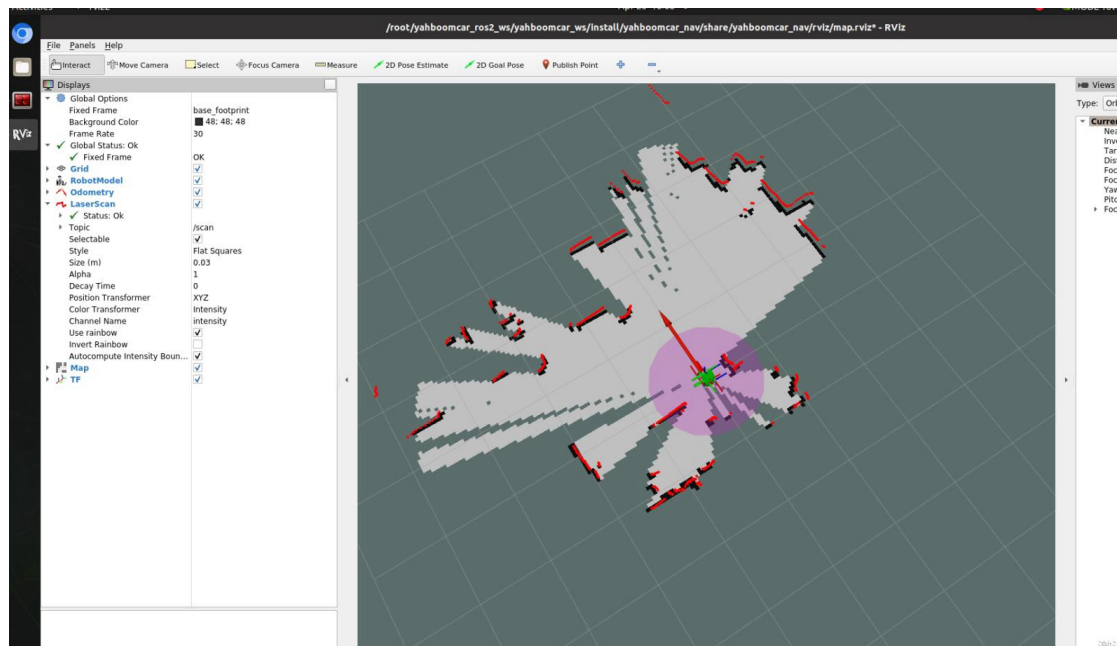
进入docker容器（步骤请参见【docker课程章节 ----- 5、进入机器人的docker容器】），分终端执行以下 launch文件：

1、 启动建图，

```
ros2 launch yahboomcar_nav map_gmapping_launch.py
```

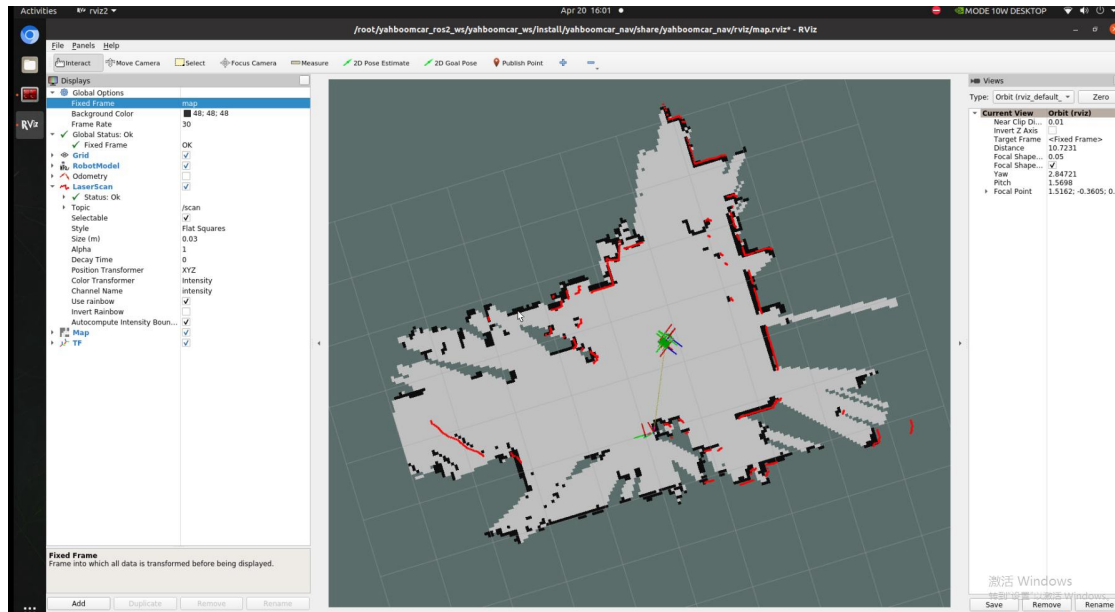
2、启动rviz显示地图，这步建议在虚拟机中执行，虚拟机中需要配置好多机通讯

```
ros2 launch yahboomcar_nav display_map_launch.py
```



3、启动键盘控制节点，这步建议在虚拟机中执行，虚拟机中需要配置好多机通讯 或者使用遥控器【缓慢移动小车】开始建图，直到建立好完整地图

```
ros2 run yahboomcar_ctrl yahboom_keyboard
```

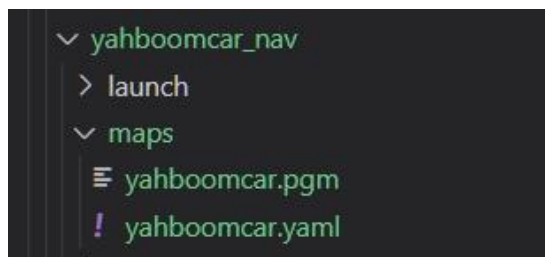


4、 地图保存，路径如下：

```
ros2 launch yahboomcar_nav save_map_launch.py
```

保存路径如下：

```
/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/
```



一个pgm图片，一个yaml文件yahboomcar.yaml

image:

```
/root/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/yahboomcar.pgm
```

mode: trinary

resolution: 0.05

origin: [-10, -10, 0]

negate: 0

occupied_thresh: 0.65

free_thresh: 0.25

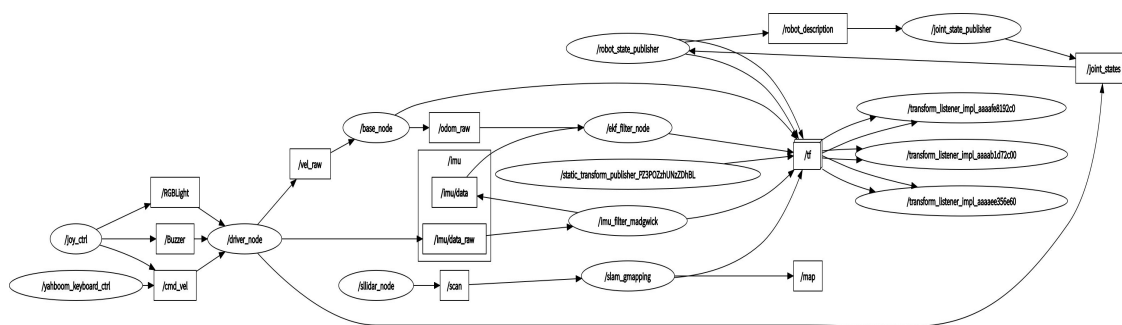
参数解析:

- image: 地图文件的路径, 可以是绝对路径, 也可以是相对路径
- mode: 该属性可以是trinary、scale或者raw之一, 取决于所选择的mode, trinary 模式是默认模式
- resolution: 地图的分辨率, 米/像素
- origin: 地图左下角的 2D 位姿(x,y,yaw), 这里的yaw是逆时针方向旋转的(yaw=0 表示没有旋转)。目前系统中的很多部分会忽略yaw值。
- negate: 是否颠倒 白/黑、自由/占用 的意义(阈值的解释不受影响)
- occupied_thresh: 占用概率大于这个阈值的像素, 会被认为是完全占用。
- free_thresh: 占用概率小于这个阈值的像素, 会被认为是完全自由。

1.3 节点解析

1.3.1 显示计算图

rqt_graph



1.3.2 gmapping 节点详情

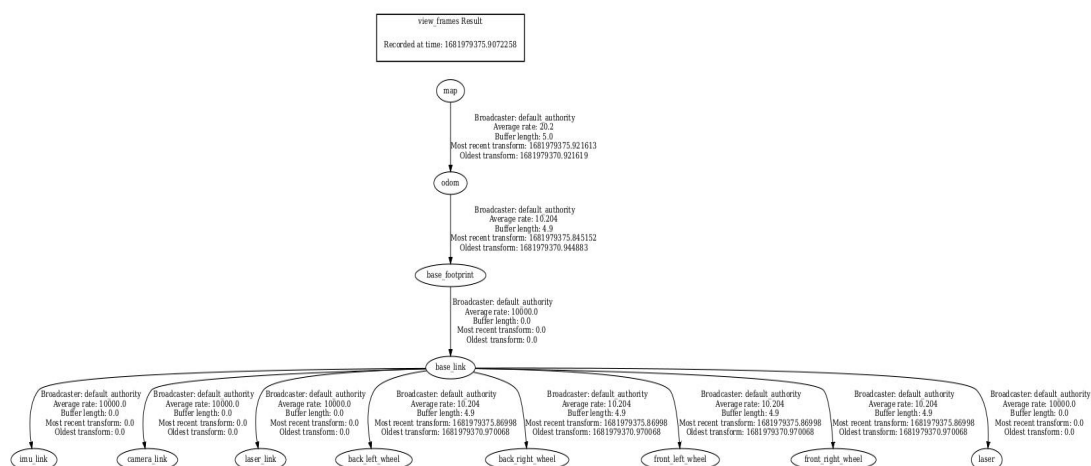
```
rr@rr-pc:~$ ros2 node info /slam_gmapping
/slam_gmapping
Subscribers:
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /scan: sensor_msgs/msg/LaserScan
Publishers:
  /entropy: std_msgs/msg/Float64
  /map: nav_msgs/msg/OccupancyGrid
  /map_metadata: nav_msgs/msg/MapMetaData
  /parameter_events: rcl_interfaces/msg/ParameterEvent
  /rosout: rcl_interfaces/msg/Log
  /tf: tf2_msgs/msg/TFMessage
Service Servers:
  /slam_gmapping/describe_parameters: rcl_interfaces/srv/DescribeParameters
  /slam_gmapping/get_parameter_types: rcl_interfaces/srv/GetParameterTypes
  /slam_gmapping/get_parameters: rcl_interfaces/srv/GetParameters
  /slam_gmapping/list_parameters: rcl_interfaces/srv/ListParameters
  /slam_gmapping/set_parameters: rcl_interfaces/srv/SetParameters
  /slam_gmapping/set_parameters_atomically: rcl_interfaces/srv/SetParametersAtomically
Service Clients:

Action Servers:

Action Clients:
```

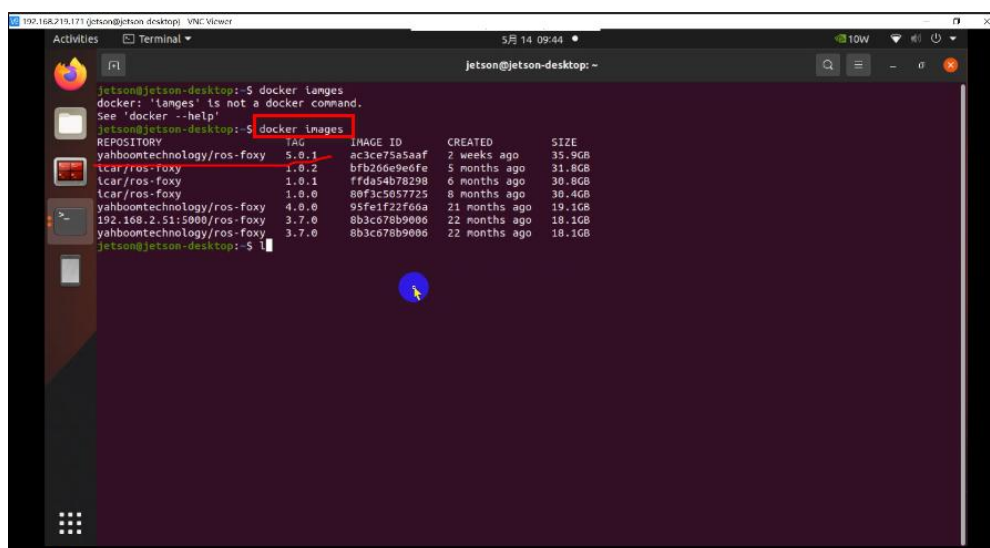
1.3.3 TF 变换

```
ros2 run tf2_tools view_frames.py
```

2 激光雷达 SLAM 建图实验过程

- 1、使用笔记本远程连接到小车系统。
- 2、在 Terminal 窗口中使用 docker images 命令查看小车系统中的镜像



如果看到上述镜像，则说明本机已经部署了建图及自动导航的 docker 镜像。

- 3、在 Terminal 窗口中使用 lsusb 指令查看 USB 设备信息

```
jetson@jetson-desktop:~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
yahboontechnology/ro- 5.0.1              ac3ce75a5aaf       2 weeks ago        35.9GB
lcar/ros-foxy        1.0.2              bfb266e9e0fe       5 months ago        31.8GB
lcar/ros-foxy        1.0.1              ffd934b78298       6 months ago        30.8GB
lcar/ros-foxy        1.0.0              88f3c5877725       8 months ago        30.4GB
yahboontechnology/ro- 4.0.0              95fe1f22f66a       21 months ago       19.1GB
192.168.2.51:5000/ro- 3.7.0              8b3c678b9006       22 months ago       18.1GB
yahboontechnology/ro- 3.7.0              8b3c678b9006       22 months ago       18.1GB

jetson@jetson-desktop:~$ lsusb
Bus 002 Device 004: ID 2109:0817 VIA Labs, Inc.
Bus 002 Device 003: ID 05e3:0626 Genesys Logic, Inc.
Bus 002 Device 002: ID 2109:0822 VIA Labs, Inc. USB3.1 Hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 003: ID 0bda:c822 Realtek Semiconductor Corp. Bluetooth Radio
Bus 001 Device 015: ID 2109:0817 VIA Labs, Inc.
Bus 001 Device 014: ID 2207:0001 Fuzhou Rockchip Electronics Company
Bus 001 Device 012: ID 8440:2004 Realtek Bluetooth Radio
Bus 001 Device 010: ID 1a86:7523 Qinheng Electronics HL-340 USB-Serial adapter
Bus 001 Device 007: ID 2109:0817 VIA Labs, Inc.
Bus 001 Device 013: ID 058f:050f VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 011: ID 2b3c:060f
Bus 001 Device 009: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 001 Device 008: ID 1a86:7523 Qinheng Electronics HL-340 USB-Serial adapter
Bus 001 Device 006: ID 05e3:0610 Genesys Logic, Inc. 4-port hub
Bus 001 Device 005: ID 10c4:e660 Silicon Labs CP210x UART Bridge
Bus 001 Device 004: ID 0d8c:0012 C-Media Electronics, Inc.
Bus 001 Device 002: ID 2109:2822 VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub

jetson@jetson-desktop:~$
```

如果看到有两行 usb 设别 ID 为 2b3c 的记录，则说明深度相机连接正常，记录下系统为其分配的设备号 011、013。

4、在 Terminal 窗口中输入指令 vi run_docker_autodrive.sh

编辑自动导航镜像的 docker 启动脚本

```
jetson@jetson-desktop:~$ vi run_docker_autodrive.sh
--net=host \
--env="DISPLAY" \
--env="QT_X11_NO_MITSHM=1" \
-v /tmp/.X11-unix:/tmp/.X11-unix \
-v /home/jetson/code/yahbooncar_ws:/root/yahbooncar_ws \
-v /home/jetson/code/software/library_ws:/root/yahbooncar_ws/software/library_ws \
-v /home/jetson/rosboard:/root/rosboard \
-v /home/jetson/maas:/root/maas \
-v /dev/bus/usb/001/013:/dev/bus/usb/001/013 \
-v /dev/bus/usb/001/011:/dev/bus/usb/001/011 \
--device=/dev/astradept \
--device=/dev/astrauvc \
--device=/dev/video0 \
--device=/dev/ttyserial \
--device=/dev/rplidar \
--device=/dev/input \
-p 9090:9090 \
-p 8888:8888 \
yahboontechnology/ro-foxy:5.0.1 /bin/bash

jetson@jetson-desktop:~$
```

将红框两行的编号更新成上一步记录的数值，比如 011、013，然后保存退出。

5、在 Terminal 串口中运行 run_docker_autodrive.sh 脚本，启动 docker 镜像容器

```

root@jetson-desktop: /
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 003: ID 0bda:c822 Realtek Semiconductor Corp. Bluetooth Radio
Bus 001 Device 015: ID 2109:0817 VIA Labs, Inc.
Bus 001 Device 014: ID 2207:0001 Fuzhou Rockchip Electronics Company
Bus 001 Device 012: ID 0440:2004 Realtek Bluetooth Radio
Bus 001 Device 010: ID 1a86:7523 QinHeng Electronics HL-340 USB-Serial adapter
Bus 001 Device 007: ID 2109:2817 VIA Labs, Inc.
Bus 001 Device 013: ID 2bc5:050f VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 011: ID 2bc5:060f
Bus 001 Device 009: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 001 Device 008: ID 1a86:7523 QinHeng Electronics HL-340 USB-Serial adapter
Bus 001 Device 006: ID 05e3:0610 Genesys Logic, Inc. 4-port hub
Bus 001 Device 005: ID 10c4:ea60 Silicon Labs CP210x UART Bridge
Bus 001 Device 004: ID 0d8c:0012 C-Media Electronics, Inc.
Bus 001 Device 002: ID 2109:2822 VIA Labs, Inc. USB2.0 Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
jetson@jetson-desktop:~$ ls
alias.txt      Desktop      Music       Rosmaster-App  software      Videos
Archtcond3-0.2.3-Linux-aarch64.sh  Documents  Pictures     run_docker_autodrive.sh  temp          vision-release-0.16
cnd.txt        Downloads   Public      run_docker.sh   Templates     yolov5-7.0
code           maps        rasboard    Samples-guide.pdf  Version.txt   Yolov5ptFile
jetson@jetson-desktop:~$ ls
alias.txt      Desktop      Music       Rosmaster-App  software      Videos
Archtcond3-0.2.3-Linux-aarch64.sh  Documents  Pictures     run_docker_autodrive.sh  temp          vision-release-0.16
cnd.txt        Downloads   Public      run_docker.sh   Templates     yolov5-7.0
code           maps        rasboard    Samples-guide.pdf  Version.txt   Yolov5ptFile
jetson@jetson-desktop:~$ ./run_docker_autodrive.sh
access control disabled, clients can connect from any host
WARNING: Published ports are discarded when using host network mode
.....
ROS_DOMAIN_ID: 30
my_robot_type: x3 | my_lidar: a1 | my_camera: astraplus
root@jetson-desktop:~#

```

6、新启动一个 Terminal 窗口，运行指令 `docker ps -a`，可以查看新启动的镜像容器

```

jetson@jetson-desktop:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
21692353449    vshboomtechnology/ros-foxy:5.0.1    "/bin/bash"            6 seconds ago Up 5 seconds          eager_hertz
a5d01aa8211    lcar/ros-foxy:1.0.2                "/bin/bash"            5 months ago  Exited (0) 5 months ago          frosty_ellon

```

记录下容器 ID 编号的前 4 位，比如这里的 2169

7、在 Terminal 窗口中输入指令 `vi .bashrc` 编辑环境变量文件

```

root@jetson-desktop: /
access control disabled, clients can connect from any host
MY_IP: 192.168.219.171
jetson@jetson-desktop:~$ docker ps -a
CONTAINER ID        IMAGE               COMMAND             CREATED             STATUS              PORTS              NAMES
2169625a3d45        yahboontechnology/ros-foxy:5.0.1  "/bin/bash"        6 seconds ago       Up 5 seconds                eager_hertz
0a50d1aa82f1        tcar/ros-foxy:1.0.2  "/bin/bash"        5 months ago        Exited (0) 5 months ago      frosty_ellon
jetson@jetson-desktop:~$ vi .bashrc

```

```

root@jetson-desktop: /
jetson@jetson-desktop:~$ cat .bashrc
if [ -f /usr/share/bash-completion/bash_completion ]; then
. /usr/share/bash-completion/bash_completion
elif [ -f /etc/bash_completion ]; then
. /etc/bash_completion
fi
alias python=python3
ip=$(ip addr show wlan0 | grep -o 'inet [0-9]\{1,\}\.[0-9]\{1,\}\.[0-9]\{1,\}\.[0-9]\{1,\}' | grep -o [0-9]\{1,\})
if [ -z $ip ]; then
ip=$(ip addr show lo | grep -o 'inet [0-9]\{1,\}\.[0-9]\{1,\}\.[0-9]\{1,\}\.[0-9]\{1,\}' | grep -o [0-9]\{1,\})
fi
echo "-----"
echo -e "MY_IP: \033[32m$ip\033[0m"
echo "-----"
export PATH=/usr/local/cuda-11.4/bin${PATH:+:${PATH}}
export LD_LIBRARY_PATH=/usr/local/cuda-11.4/lib64${LD_LIBRARY_PATH:+:${LD_LIBRARY_PATH}}
export CUDA_ROOT=/usr/local/cuda

alias ros='cd /home/jetson/Rosmaster-App/rosmaster/'
alias run='/home/jetson/Rosmaster-App/rosmaster/start_app.sh'
alias start='python rosmaster_pid.py 50 40 2000'
alias stop='python rosmaster_pid.py 50 40 2000'
alias app='cd /home/jetson/Rosmaster-App/rosmaster;python3 app_sin_run.py'
alias back='if [ ! -d ~/.valid ]; then mkdir -p ~/.valid;fi;cp -f /Rosmaster-App/rosmaster/license.json ~/.valid;cp -f /Rosmaster-App/rosmaster/enc.txt ~/.valid'
alias restore='cp -f ~/.valid/license.json ~/Rosmaster-App/rosmaster/;cp -f ~/.valid/enc.txt ~/Rosmaster-App/rosmaster/'

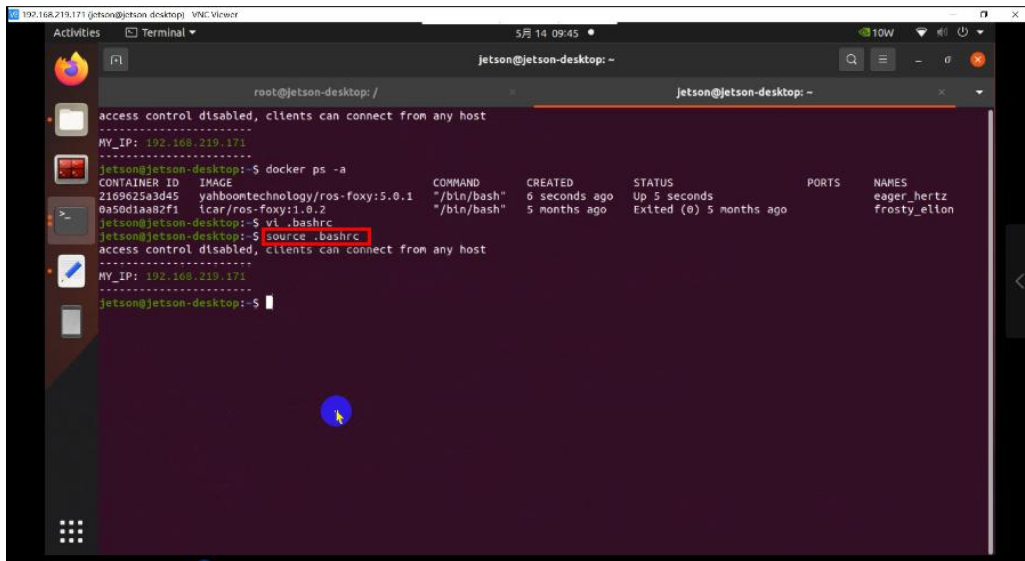
alias s='docker start 2169'
alias t='docker stop 2169'
alias d='docker exec -it 2169 /bin/bash'

```

编辑红框的几行，更新容器 ID 为刚刚记录的值，比如:2169。

保存退出。

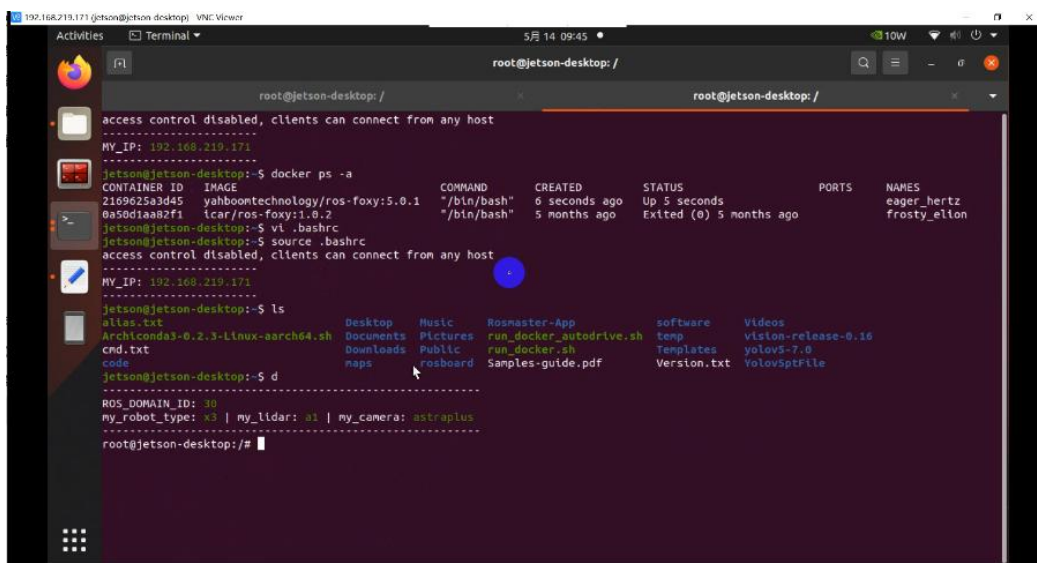
8、在 Terminal 窗口中执行指令 `source .bashrc` 即可生效 s、t、d 单个快捷指令



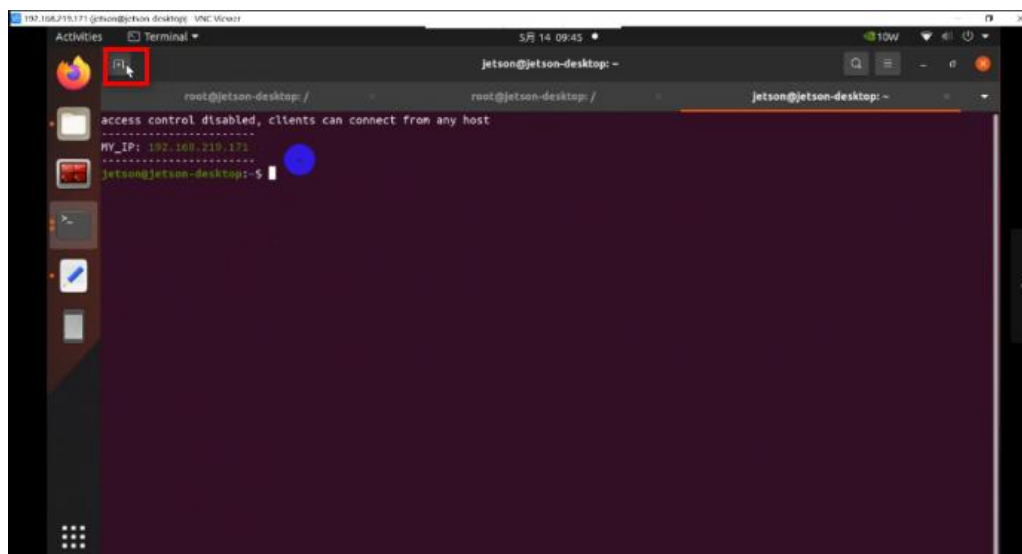
此时，在 Terminal 窗口中可以通过 s、t、d 三个快捷指令实现对自动驾驶容器的启动、停止和登入操作：

快捷指令	完整指令	指令用途
s	docker start 2169	启动 ID 前缀为 2169 的 docker 容器
t	docker stop 2169	停止 ID 前缀为 2169 的 docker 容器
d	docker exec -it 2169 /bin/bash	登入 ID 前缀为 2169 的 docker 容器

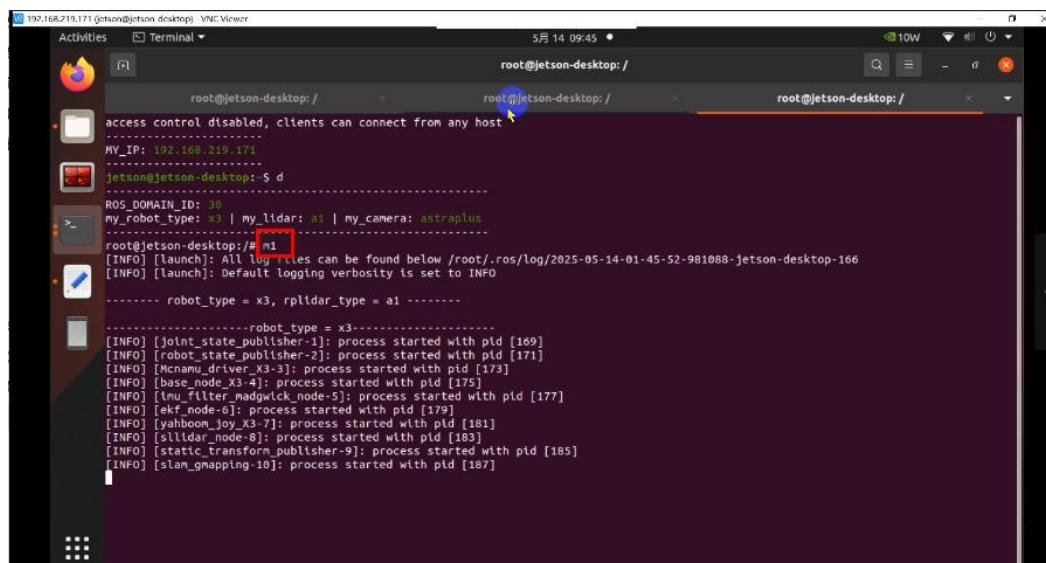
9、新启动两个 Terminal 窗口，输入快捷指令 d，登入 docker 容器环境备用



其中，新启动 Terminal 窗口可以通过点击左上角的+号：



10、在其中一个 docker 容器环境中输入指令 m1，启动建图时的基础应用 ROS 节点



激光 SLAM 建图涉及 4 个指令，分别如下：

快捷指令	完整指令	指令用途
m1	ros2 launch yahboomcar_nav map_gmapping_launch.py	启动 gmapping 建图的基础节点
m2	ros2 launch yahboomcar_nav	启动 gmapping 建图的地图展

	display_map_launch.py	示
m3	ros2 run yahboomcar_ctrl yahboom_keyboard	启动键盘控制节点, 控制小车运动
m4	ros2 launch yahboomcar_nav save_map_launch.py	保存地图

这 4 个指令的定义在 docker 容器环境的/root/.bashrc 文件中。

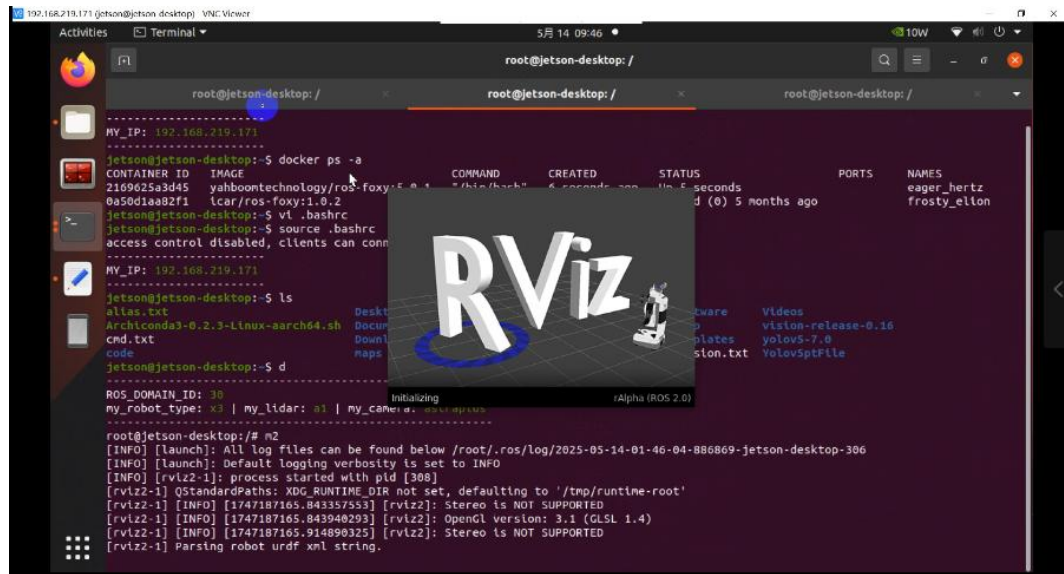
11、在其中一个 docker 容器环境中输入指令 m2，启动建图时的地图展示

```

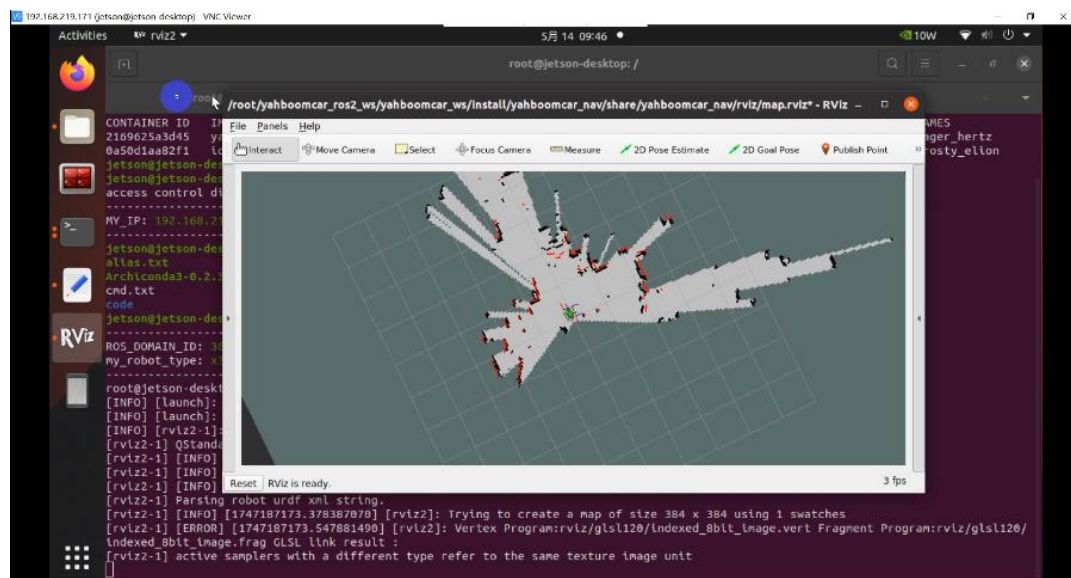
root@jetson-desktop: /
root@jetson-desktop: /
root@jetson-desktop: /
access control disabled, clients can connect from any host
MY_IP: 192.168.219.171
jetson@jetson-desktop:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
21696253d45   yahboomtechnology/ros-foxy:5.0.1   /bin/bash               6 seconds ago Up 5 seconds          eager_hertz
0a50d1aa2f1    icar/ros-foxy:1.0.2               /bin/bash               5 months ago  Exited (0) 5 months ago          frosty_ellon
jetson@jetson-desktop:~$ vi .bashrc
jetson@jetson-desktop:~$ source .bashrc
access control disabled, clients can connect from any host
MY_IP: 192.168.219.171
jetson@jetson-desktop:~$ ls
alias.txt      Desktop      Music        Rosmaster-App  software      Videos
Archlinuxda3-0.2.3-Linux-aarch64.sh Documents    Pictures      run_docker_autodrive.sh  temp          vision-release-0.16
cnd.txt        Downloads   Public       run_docker.sh   Templates     yolov5-7.0
code           maps        rosboard    Samples-guide.pdf  Version.txt   Yolov5ptFile
jetson@jetson-desktop:~$ d
-----
ROS_DOMAIN_ID: 30
my_robot_type: x3 | my_lidar: a1 | my_camera: astraplus
-----
root@jetson-desktop:~# m2
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-05-14-01-46-04-886869-jetson-desktop-306
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [rviz2-1]: process started with pid [308]

```

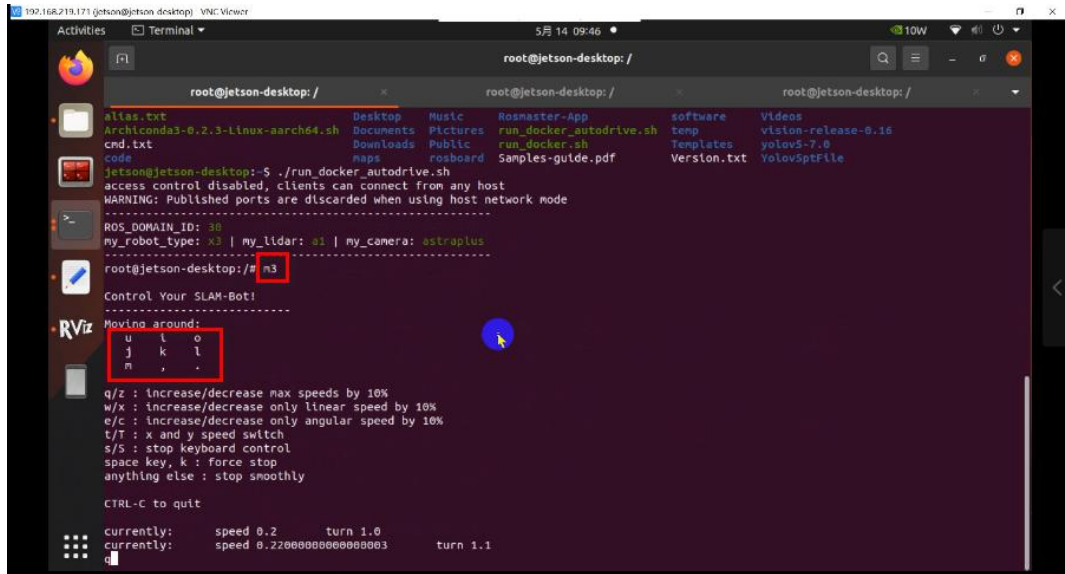
该指令将启动 RViz:



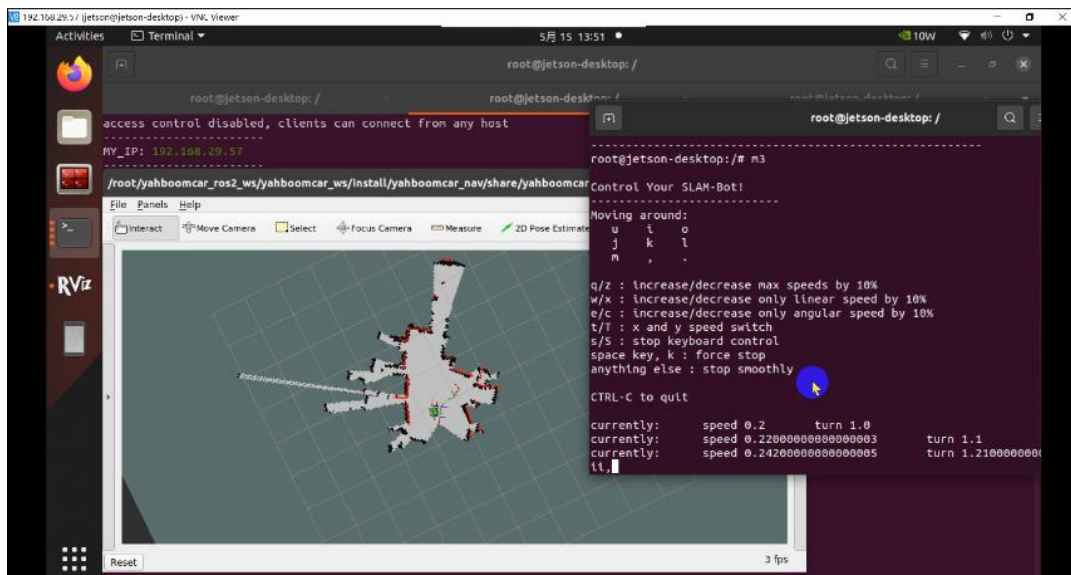
实时展示当前激光扫描到的地图状况：



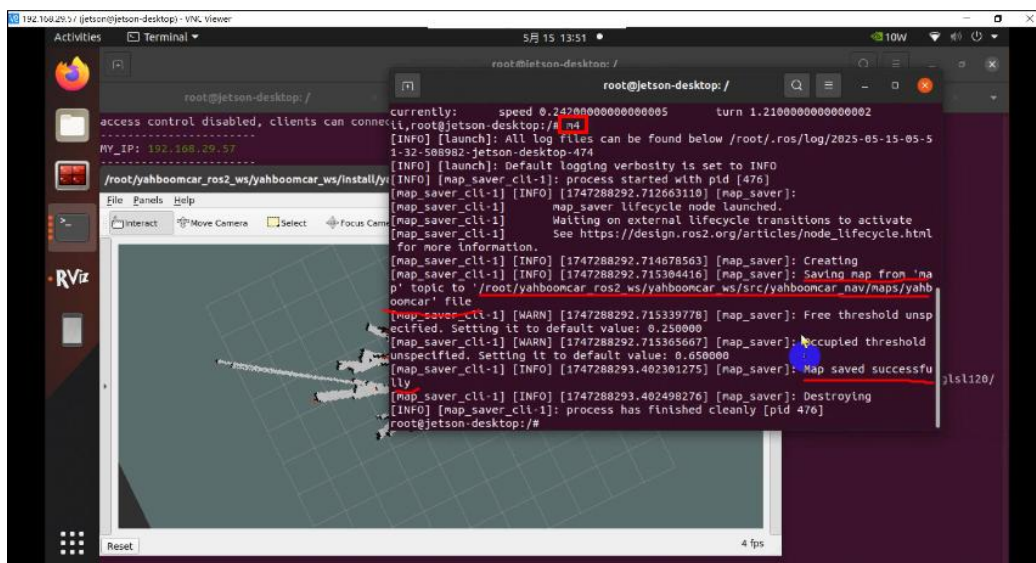
12、在其中一个 docker 容器环境中输入指令 m3，启动键盘控制小车运动进行建图



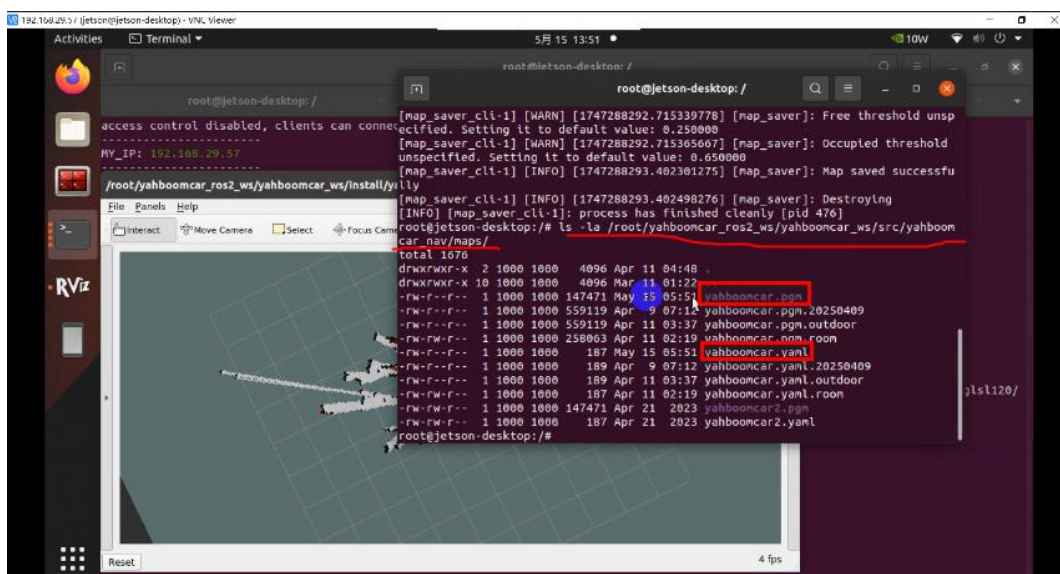
通过键盘的 'u' 、 'i' 、 'o' 、 'j' 、 'k' 、 'l' 、 'm' 、 ',' 、 '.' 等按键控制
小车在空间中移动，逐步扫描过测试空间，构建出完整地图：



13、建图完成后，用 ctrl+c 退出 m3 运行，输入 m4 指令保存地图：

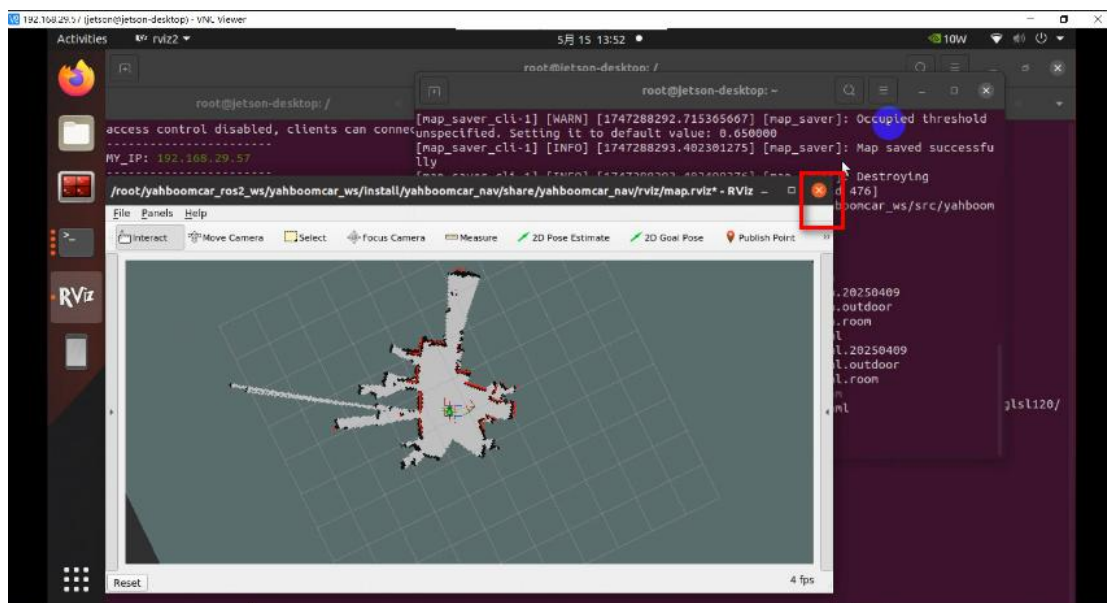


此时如果查看保存路径，可以看到刚刚保存的地图信息文件：

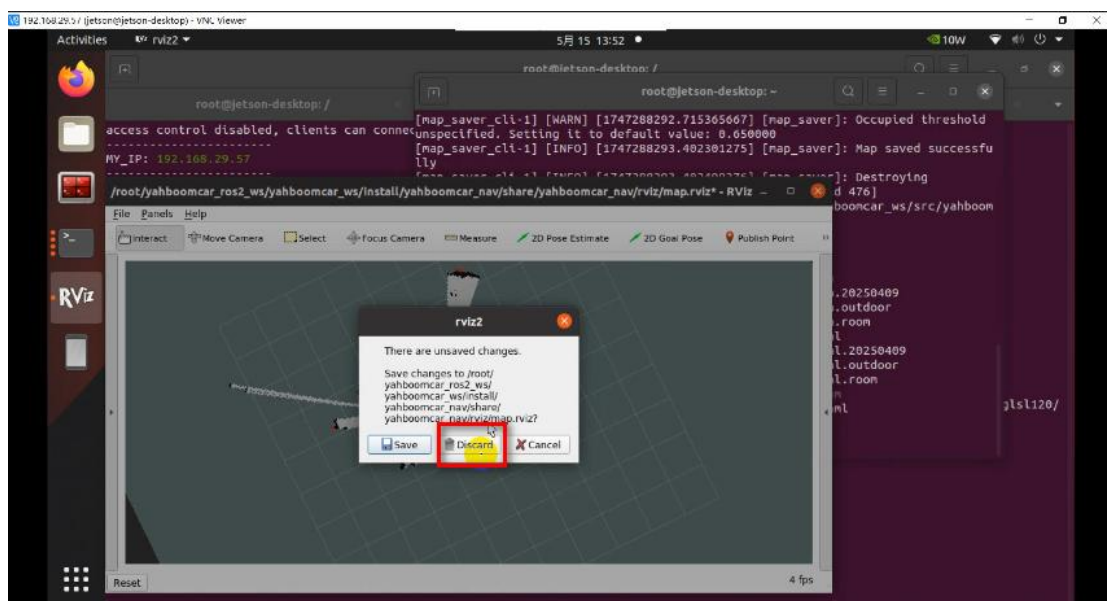


14、停止 m1 和 m2 的指令运行。

其中停止 m2 执行，通过点击 RViz 的右上角的红叉：



在弹出对话框中选择“Discard”即可。



15、停止 m1 指令的运行在其窗口被激活的状态，按 ctrl + c 即可

```

root@jetson-desktop: /
root@jetson-desktop: /
root@jetson-desktop: /

[Mcnamu_driver_X3-3] File "/usr/local/lib/python3.8/dist-packages/Rosmaster_Lib-3.3.1-py3.8.egg/Rosmaster_Lib/Rosmaster_Lib.py", line 1212, in get_version
[Mcnamu_driver_X3-3] KeyboardInterrupt
[yahboom_joy_X3-7] Traceback (most recent call last):
[yahboom_joy_X3-7] File "/root/yahboomcar_ros2_ws/yahboomcar_ws/install/yahboomcar_ctrl/lib/yahboomcar_ctrl/yahboom_joy_X3", line 11, in <module>
[yahboom_joy_X3-7]   load_entry_point('yahboomcar-ctrl=0.0.0', 'console_scripts', 'yahboom_joy_X3')()
[yahboom_joy_X3-7] File "/root/yahboomcar_ros2_ws/yahboomcar_ws/install/yahboomcar_ctrl/lib/python3.8/site-packages/yahboomcar_ctrl/yahboom_joy_X3.py", line 158, in main
[yahboom_joy_X3-7]   rclpy.spin(joy_ctrl)
[yahboom_joy_X3-7] File "/opt/ros/foxy/lib/python3.8/site-packages/rclpy/__init__.py", line 191, in spin
[yahboom_joy_X3-7]   executor.spin_once()
[yahboom_joy_X3-7] File "/opt/ros/foxy/lib/python3.8/site-packages/rclpy/executors.py", line 710, in spin_once
[yahboom_joy_X3-7]   handler, entity, node = self.wait_for_ready_callbacks(timeout_sec=timeout_sec)
[yahboom_joy_X3-7] File "/opt/ros/foxy/lib/python3.8/site-packages/rclpy/executors.py", line 696, in wait_for_ready_callbacks
[yahboom_joy_X3-7]   return next(self._cb_iter)
[yahboom_joy_X3-7] File "/opt/ros/foxy/lib/python3.8/site-packages/rclpy/executors.py", line 593, in _wait_for_ready_callbacks
[yahboom_joy_X3-7]   _rclpy.rclpy_wait(wait_set, timeout_nsec)
[yahboom_joy_X3-7] KeyboardInterrupt
[INFO] [imu_filter_madgwick_node-5]: process has finished cleanly [pid 232]
[INFO] [static_transform_publisher-9]: process has finished cleanly [pid 240]
[INFO] [base_node_X3-4]: process has finished cleanly [pid 230]
[INFO] [robot_state_publisher-2]: process has finished cleanly [pid 226]
[INFO] [slam_gmapping-10]: process has finished cleanly [pid 242]
[INFO] [ekf_node-6]: process has finished cleanly [pid 234]
[INFO] [joint_state_publisher-1]: process has finished cleanly [pid 224]
[ERROR] [yahboom_joy_X3-7]: process has died [pid 236, exit code -2, cmd '/root/yahboomcar_ros2_ws/yahboomcar_ws/install/yahboomcar_ctrl/lib/yahboomcar_ctrl/yahboom_joy_X3 --ros-args'].
[ERROR] [Mcnamu_driver_X3-3]: process has died [pid 228, exit code -2, cmd '/root/yahboomcar_ros2_ws/yahboomcar_ws/install/yahboomcar_ctrl/lib/yahboomcar_ctrl/yahboom_joy_X3 --ros-args'].
[sllidar_node-8] [INFO] [1747288359.221719252] [sllidar_node]: Stop motor
[INFO] [sllidar_node-8]: process has finished cleanly [pid 238]
root@jetson-desktop: /#

```